

Computer Networks

Assignment 1

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Part 1.

1. .

IP address = 10.5.16.38

Subnet Mask = 255.255.255.0

Network ID = 10.5.16.0

```
user@user-Veriton-S2690G-D22E2:~$ ifconfig
enp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.5.16.38 netmask 255.255.255.0 broadcast 10.5.16.255
    inet6 fe80::8aae:ddff:fe33:a0b6 prefixlen 64 scopeid 0x20<link>
    ether 88:ae:dd:33:a0:b6 txqueuelen 1000 (Ethernet)
    RX packets 53435 bytes 9790272 (9.7 MB)
    RX errors 0 dropped 345 overruns 0 frame 0
    TX packets 1265 bytes 130622 (130.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1836 bytes 572857 (572.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1836 bytes 572857 (572.8 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. .

We can see that for different CDN servers there are different IP addresses for the same domain. This is done for various reasons such as load balancing, lowering geographical latency. This technique allows the request to be routed to the closest server hosting the given website.

```

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 157.240.13.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f188:84:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.164
Server:          172.16.1.164
Address:         172.16.1.164#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 157.240.16.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f188:84:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.165
Server:          172.16.1.165
Address:         172.16.1.165#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.166
Server:          172.16.1.166
Address:         172.16.1.166#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 31.13.79.35
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f12f:183:face:b00c:0:25de

user@user-Veriton-S2690G-D22E2:~$ nslookup www.facebook.com 172.16.1.180
Server:          172.16.1.180
Address:         172.16.1.180#53

Non-authoritative answer:
www.facebook.com canonical name = star-mini.c10r.facebook.com.
Name:   star-mini.c10r.facebook.com
Address: 57.144.124.1
Name:   star-mini.c10r.facebook.com
Address: 2a03:2880:f33e:8:face:b00c:0:25de

```

3. .

In the following screenshot, minimum RTT, average RTT, maximum RTT and packet loss can be seen for different packet sizes. The packets have additional 28 bytes of headers included over their base size.

```

user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 64 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 64(92) bytes of data.
 72 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.366 ms
 72 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.355 ms
 72 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.351 ms
 72 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.322 ms
 72 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.480 ms
 72 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.490 ms
 72 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.488 ms
 72 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.487 ms
 72 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.358 ms
 72 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.349 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9223ms
rtt min/avg/max/mdev = 0.322/0.404/0.490/0.067 ms
user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 128 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 128(156) bytes of data.
136 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.212 ms
136 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.378 ms
136 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.370 ms
136 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.364 ms
136 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.491 ms
136 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.300 ms
136 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.480 ms
136 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.504 ms
136 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.488 ms
136 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.467 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9224ms
rtt min/avg/max/mdev = 0.212/0.405/0.504/0.092 ms
user@user-Veriton-S2690G-D22E2:~$ ping -c 10 -s 512 -W 100 10.5.16.37
PING 10.5.16.37 (10.5.16.37) 512(540) bytes of data.
520 bytes from 10.5.16.37: icmp_seq=1 ttl=64 time=0.252 ms
520 bytes from 10.5.16.37: icmp_seq=2 ttl=64 time=0.388 ms
520 bytes from 10.5.16.37: icmp_seq=3 ttl=64 time=0.372 ms
520 bytes from 10.5.16.37: icmp_seq=4 ttl=64 time=0.375 ms
520 bytes from 10.5.16.37: icmp_seq=5 ttl=64 time=0.387 ms
520 bytes from 10.5.16.37: icmp_seq=6 ttl=64 time=0.394 ms
520 bytes from 10.5.16.37: icmp_seq=7 ttl=64 time=0.379 ms
520 bytes from 10.5.16.37: icmp_seq=8 ttl=64 time=0.388 ms
520 bytes from 10.5.16.37: icmp_seq=9 ttl=64 time=0.372 ms
520 bytes from 10.5.16.37: icmp_seq=10 ttl=64 time=0.391 ms

--- 10.5.16.37 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9198ms
rtt min/avg/max/mdev = 0.252/0.369/0.394/0.040 ms

```

4. .

Number of hops - **12**

Number of visible hosts - **13** (upper bound of $12 * 3 = 36$)

In some hops *** is seen, which represents that the intermediate host did not respond. This can be caused due to multiple reasons such as router overload, firewall, port blocking

```

user@user-Veriton-526906-022E2:~$ traceroute www.google.com
traceroute to www.google.com (142.250.183.164), 30 hops max, 60 byte packets
 1 _gateway (10.5.16.2)  0.518 ms  0.483 ms  0.470 ms
 2 10.120.2.33 (10.120.2.33)  0.457 ms  0.443 ms  0.431 ms
 3 10.255.1.3 (10.255.1.3)  3.828 ms  3.828 ms  3.920 ms
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 72.14.204.62 (72.14.204.62)  37.892 ms  46.059 ms  46.547 ms
 9 * * *
10 142.250.214.102 (142.250.214.102)  56.496 ms  142.251.77.100 (142.251.77.100)  39.879 ms  172.253.77.22 (172.253.77.22)  40.906 ms
11 142.251.64.15 (142.251.64.15)  46.545 ms  142.251.64.13 (142.251.64.13)  46.456 ms  142.251.64.15 (142.251.64.15)  49.012 ms
12 142.250.208.227 (142.250.208.227)  58.509 ms  bom07s32-lin-r4.1e100.net (142.250.183.164)  46.782 ms  142.250.226.67 (142.250.226.67)  46.664 ms

```

Part 2.

1. Analysis of DNS Packets: Structure and its Traffic

- DNS uses UDP for packet transmission
- IP of source machine - 192.168.137.19, IP of destination machine - 192.168.137.1
- 2 DNS queries were sent to the DNS server
- 1 server responded with actual IP address in query for A type record.
- Only 1 DNS server was involved which responded.

```

parth@parth-HP-Laptop-15s-fr1xxx:~$ ( nmcli dev list || nmcli dev show ) 2>/dev/
null | grep DNS
IP4.DNS[1]: 192.168.137.1

```

f. .

*wlo1						
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
dns.qry.name == "www.iitkgp.ac.in"						
No.	Time	Source	Destination	Protocol	Length	Info
151	17.482512819	192.168.137.19	192.168.137.1	DNS	76	Standard query 0x0692 A www.iitkgp.ac.in
152	17.482755709	192.168.137.19	192.168.137.1	DNS	76	Standard query 0x752a HTTPS www.iitkgp.ac.in
158	17.495420768	192.168.137.1	192.168.137.19	DNS	108	Standard query response 0x0692 A www.iitkgp.ac.in
167	17.502578845	192.168.137.1	192.168.137.19	DNS	76	Standard query response 0x752a No such name HTTP

Queries

- www.iitkgp.ac.in: type A, class IN
 - Name: www.iitkgp.ac.in
 - [Name Length: 16]
 - [Label Count: 4]
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)

Answers

- www.iitkgp.ac.in: type A, class IN, addr 172.16.3.10
 - Name: www.iitkgp.ac.in
 - Type: A (1) (Host Address)
 - Class: IN (0x0001)
 - Time to live: 0 (0 seconds)
 - Data length: 4
 - Address: 172.16.3.10

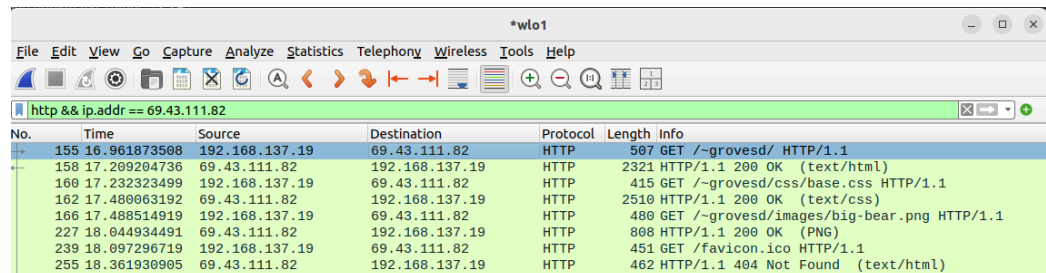
[Request In: 151]
[Time: 0.012907949 seconds]

wireshark_wlo1F95B02.pcapng

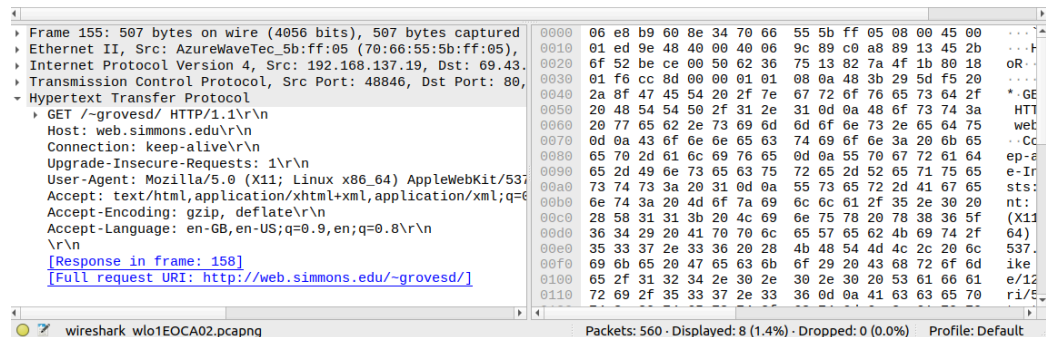
Packets: 19508 · Displayed: 4 (0.0%) · Dropped: 0 (0.0%) · Profile: Default

2. Web Traffic (HTTP)

a. .



No.	Time	Source	Destination	Protocol	Length	Info
155	16.961873508	192.168.137.19	69.43.111.82	HTTP	507	GET /~grovesd/ HTTP/1.1
158	17.209294736	69.43.111.82	192.168.137.19	HTTP	2321	HTTP/1.1 200 OK (text/html)
160	17.232323499	192.168.137.19	69.43.111.82	HTTP	415	GET /~grovesd/css/base.css HTTP/1.1
162	17.480063192	69.43.111.82	192.168.137.19	HTTP	2510	HTTP/1.1 200 OK (text/css)
166	17.488514919	192.168.137.19	69.43.111.82	HTTP	480	GET /~grovesd/images/big-bear.png HTTP/1.1
227	18.044934491	69.43.111.82	192.168.137.19	HTTP	808	HTTP/1.1 200 OK (PNG)
239	18.097296719	192.168.137.19	69.43.111.82	HTTP	451	GET /favicon.ico HTTP/1.1
255	18.361930905	69.43.111.82	192.168.137.19	HTTP	462	HTTP/1.1 404 Not Found (text/html)



Frame 155: 507 bytes on wire (4056 bits), 507 bytes captured	0000	06 e8 b9 60 8e 34 70 66	55 5b ff 05 08 00 45 00	...
↳ Ethernet II, Src: AzureWaveTec_5b:ff:05 (70:66:55:5b:ff:05),	0010	01 ed 9e 48 40 00 40 06	9c 89 c0 a8 89 13 45 2b	...
↳ Internet Protocol Version 4, Src: 192.168.137.19, Dst: 69.43.	0020	0f 52 be ce 00 50 62 36	75 13 82 7a 4f 1b 80 18	oR...
↳ Transmission Control Protocol, Src Port: 48846, Dst Port: 80,	0030	01 f6 cc 8d 00 00 01 01	08 0a 48 3b 29 5d f5 20	...
↳ Hypertext Transfer Protocol	0040	2a 8f 47 45 54 20 2f 7e	67 72 6f 76 65 73 64 2f	*.GE
↳ GET /~grovesd/ HTTP/1.1\r\n	0050	20 48 54 54 50 2f 31 2e	31 0d 0a 48 6f 73 74 3a	HTT
Host: web.simmons.edu\r\n	0060	20 77 65 62 2e 73 69 6d	6d 6f 6e 73 2e 65 64 75	web
Connection: keep-alive\r\n	0070	0d 0a 43 6f 6e 6e 65 63	74 69 6f 6e 3a 20 6b 65	..Cc
Upgrade-Insecure-Requests: 1\r\n	0080	65 70 2d 61 6c 69 76 65	0d 0a 55 70 67 72 61 64	ep-a
User-Agent: Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537	0090	65 2d 49 6e 73 65 63 75	72 65 2d 52 65 71 75 65	e-ir
Accept: text/html,application/xhtml+xml,application/xml;q=0.9	00a0	73 74 73 3a 20 31 0d 0a	55 73 65 72 2d 41 67 65	sts:
Accept-Encoding: gzip, deflate\r\n	00b0	6e 74 3a 20 4d 6f 7a 69	0c 6c 61 2f 35 2e 30 20	nt:
Accept-Language: en-GB,en-US;q=0.9,en;q=0.8\r\n	00c0	28 58 31 31 3b 20 4c 69	6e 75 78 20 78 38 36 5f	(X11
\r\n	00d0	36 34 29 20 41 70 70 6c	65 57 65 62 4b 69 74 2f	64)
[Response in frame: 158]	00e0	35 33 37 2e 33 36 20 28	4b 48 54 4d 4c 2c 20 6c	537.
[Full request URI: http://web.simmons.edu/~grovesd/]	00f0	69 6b 65 20 47 65 63 6b	6f 29 20 43 68 72 6f 6d	ike
	0100	65 2f 31 32 34 2e 30 2e	30 2e 30 20 53 61 66 61	e/12
	0110	72 69 2f 35 33 37 2e 33	36 0d 0a 41 63 63 65 70	ri/5

- b. Header of requests and responses have different format which can be analyzed to see which packet belongs to request and which to response
- HTTP request headers contain fields like Request Method, Request URI, Request Version, Host, User-Agent, Referrer etc.
- HTTP response headers have fields like Response Version, Status Code, Response Phrase, Date, Content-Type etc.
- c. 8 packets are exchanged between client (browser) and server (69.43.111.82) to load the entire web page

3. ICMP Traffic (Ping/Traceroute)

a. .

Source of IP packets - 10.145.156.238

Destination of packets - 10.145.22.3

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ ping 10.145.22.3
PING 10.145.22.3 (10.145.22.3) 56(84) bytes of data.
64 bytes from 10.145.22.3: icmp_seq=1 ttl=61 time=319 ms
64 bytes from 10.145.22.3: icmp_seq=2 ttl=61 time=148 ms
64 bytes from 10.145.22.3: icmp_seq=3 ttl=61 time=161 ms
64 bytes from 10.145.22.3: icmp_seq=4 ttl=61 time=184 ms
64 bytes from 10.145.22.3: icmp_seq=5 ttl=61 time=109 ms
64 bytes from 10.145.22.3: icmp_seq=6 ttl=61 time=244 ms
64 bytes from 10.145.22.3: icmp_seq=7 ttl=61 time=455 ms
64 bytes from 10.145.22.3: icmp_seq=8 ttl=61 time=180 ms
64 bytes from 10.145.22.3: icmp_seq=9 ttl=61 time=195 ms
64 bytes from 10.145.22.3: icmp_seq=10 ttl=61 time=217 ms
64 bytes from 10.145.22.3: icmp_seq=11 ttl=61 time=447 ms
^C
--- 10.145.22.3 ping statistics ---
12 packets transmitted, 11 received, 8.33333% packet loss, time 11011ms
rtt min/avg/max/mdev = 109.063/241.787/455.192/111.360 ms
```

3	9.611706	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
4	9.930639	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
5	10.612696	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
6	10.761106	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
7	11.614328	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
8	11.775514	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
9	12.615729	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
10	12.799641	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
11	13.616822	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
12	13.725853	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
13	14.618122	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
14	14.849380	AzureWaveTec_5b:ff:...	Fortinet_09:01:1e	ARP	42 Who has 10.145.128.2?
15	14.861923	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
16	14.861923	Fortinet_09:01:1e	AzureWaveTec_5b:ff:...	ARP	56 10.145.128.2 is at 00
17	15.619451	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
18	16.074603	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
19	16.619513	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
20	16.799165	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
21	17.621343	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request
22	17.816382	10.145.22.3	10.145.156.238	ICMP	98 Echo (ping) reply
23	18.621776	10.145.156.238	10.145.22.3	ICMP	98 Echo (ping) request

3 packets have TTL = 1 and 3 have TTL = 2, which is the expected behaviour according to tracerouting algorithm.

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ traceroute 10.145.22.3
traceroute to 10.145.22.3 (10.145.22.3), 30 hops max, 60 byte packets
1 _gateway (10.145.128.2) 11.581 ms 11.516 ms 11.490 ms
2 10.250.1.4 (10.250.1.4) 11.473 ms 11.456 ms 11.440 ms
3 10.120.0.26 (10.120.0.26) 11.421 ms 11.753 ms 11.388 ms
4 10.145.22.3 (10.145.22.3) 494.283 ms 494.266 ms 494.249 ms
```


31	30.837534	10.145.156.238	10.145.22.3	UDP	74 58007 → 33434	Len=32
32	30.837573	10.145.156.238	10.145.22.3	UDP	74 37319 → 33435	Len=32
33	30.837598	10.145.156.238	10.145.22.3	UDP	74 38176 → 33436	Len=32
34	30.837615	10.145.156.238	10.145.22.3	UDP	74 53682 → 33437	Len=32
35	30.837632	10.145.156.238	10.145.22.3	UDP	74 38247 → 33438	Len=32
36	30.837649	10.145.156.238	10.145.22.3	UDP	74 36148 → 33439	Len=32
37	30.837666	10.145.156.238	10.145.22.3	UDP	74 53858 → 33440	Len=32
38	30.837682	10.145.156.238	10.145.22.3	UDP	74 59277 → 33441	Len=32
39	30.837699	10.145.156.238	10.145.22.3	UDP	74 49703 → 33442	Len=32
40	30.837717	10.145.156.238	10.145.22.3	UDP	74 39663 → 33443	Len=32
41	30.837733	10.145.156.238	10.145.22.3	UDP	74 55777 → 33444	Len=32
42	30.837750	10.145.156.238	10.145.22.3	UDP	74 41254 → 33445	Len=32
43	30.837769	10.145.156.238	10.145.22.3	UDP	74 36949 → 33446	Len=32
44	30.837786	10.145.156.238	10.145.22.3	UDP	74 58464 → 33447	Len=32
45	30.837803	10.145.156.238	10.145.22.3	UDP	74 35762 → 33448	Len=32
46	30.837820	10.145.156.238	10.145.22.3	UDP	74 56350 → 33449	Len=32
47	30.849083	10.145.128.2	10.145.156.238	ICMP	102 Time-to-live exceeded	
48	30.849083	10.145.128.2	10.145.156.238	ICMP	102 Time-to-live exceeded	
49	30.849083	10.145.128.2	10.145.156.238	ICMP	102 Time-to-live exceeded	
50	30.849083	10.120.0.26	10.145.156.238	ICMP	70 Time-to-live exceeded	
51	30.849083	10.120.0.26	10.145.156.238	ICMP	70 Time-to-live exceeded	
52	30.849084	10.250.1.4	10.145.156.238	ICMP	70 Time-to-live exceeded	
53	30.849084	10.250.1.4	10.145.156.238	ICMP	70 Time-to-live exceeded	
54	30.849084	10.250.1.4	10.145.156.238	ICMP	70 Time-to-live exceeded	
55	30.849431	10.120.0.26	10.145.156.238	ICMP	70 Time-to-live exceeded	

b. .

Sent ICMP packets receive no response.

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ ping 192.168.31.3
PING 192.168.31.3 (192.168.31.3) 56(84) bytes of data.
^C
--- 192.168.31.3 ping statistics ---
11 packets transmitted, 0 received, 100% packet loss, time 10278ms
```

91	60.314177	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
92	61.376616	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
93	62.400615	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
94	63.424529	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
95	64.448591	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
98	65.472524	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
101	66.496581	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
102	67.520583	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
103	68.544588	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
104	69.568591	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request
105	70.592597	10.145.156.238	192.168.31.3	ICMP	98 Echo (ping) request

c. .

While trying to trace an unreachable host, the first 6 hops are being traced and a packet is being returned. But after that there is no packet reply, hence it can be assumed that there is no viable route to reach the destination address. (Default max. number of hops is 30)

```
parth@parth-HP-Laptop-15s-fr1xxx:~$ traceroute 192.168.31.3
traceroute to 192.168.31.3 (192.168.31.3), 30 hops max, 60 byte packets
 1 _gateway (10.145.128.2) 9.714 ms 9.645 ms 8.560 ms
 2 192.168.255.18 (192.168.255.18) 8.538 ms 8.521 ms 8.862 ms
 3 10.119.228.129 (10.119.228.129) 8.476 ms 8.460 ms 8.443 ms
 4 10.173.35.1 (10.173.35.1) 38.255 ms 10.173.35.253 (10.173.35.253) 107.832 ms 107.810 ms
 5 10.255.239.189 (10.255.239.189) 107.794 ms 10.255.238.166 (10.255.238.166) 38.183 ms 10.255.239.189 (10.255.239.189) 107.760 ms
 6 10.152.7.38 (10.152.7.38) 107.741 ms 93.061 ms 92.993 ms
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